# COUNTING BRANCHES AT A POINT OF A DECENT SPACE: FOUR EQUIVALENT FORMULATIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 5736-5746, immediately following the statement at 5716-5734.

*Editorial note.* Source line 5740 reads 'The ring O_{X,x} is the henselization of O_{U,u}' without the superscript h on O_{X,x}; this appears to be a typo for the henselian local ring O^h_{X,x} (compare lemma-irreducible-local-ring, source line 5694). The author text is retained verbatim and this note flags the reading.

*Editorial note.* Two cross-chapter pointers (More on Algebra, More on Morphisms) are recorded in assembly.external_references.

*Allowed standard prerequisites (editorial).*

- Minimal primes versus irreducible components through a point. standard commutative-algebra correspondence invoked at proof lines 5736-5738. Carries the equivalence (1) <=> (2) with counts.
- The henselian local ring is the henselization of the chart. discussion following Definition definition-henselian-local-ring of this chapter, cited at proof lines 5739-5741. Bridges clause (4) (about O^h_{X,x}) with the chart-side clauses.
- Unibranch characterization via henselization. More on Algebra, Lemma more-algebra-lemma-unibranch, cited at proof lines 5742-5743. Carries the equivalence (3) <=> (4): the number of branches equals the number of minimal primes of the henselization.
- Branches count via henselization. More on Morphisms, Lemma more-morphisms-lemma-nr-branches, cited at proof lines 5744-5746. Carries the equivalence (2) <=> (3).

*Source reference.* Original definition 0BGW: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2716–2716. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0BQ0: focused-seeds/geometry/S9-decent-nr-branches-local-ring/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0CB4: focused-seeds/geometry/S9-decent-nr-branches-local-ring/ref-tags.txt, source lines 6–6. Navigation pointer only: the

*Date*: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be a decent algebraic space over $S$. Let $x \in |X|$ be a point. Let $n \in \{1, 2, \ldots\}$ be an integer. The following are equivalent*

1. *for any elementary étale neighbourhood $(U, u) \to (X, x)$ the number of minimal primes of the local ring $\mathcal{O}_{U,u}$ is $\leq n$ and for at least one choice of $(U, u)$ it is $n$,*
2. *for any elementary étale neighbourhood $(U, u) \to (X, x)$ the number irreducible components of $U$ passing through $u$ is $\leq n$ and for at least one choice of $(U, u)$ it is $n$,*
3. *for any elementary étale neighbourhood $(U, u) \to (X, x)$ the number of branches of $U$ at $u$ is $\leq n$ and for at least one choice of $(U, u)$ it is $n$,*
4. *the number of minimal prime ideals of $\mathcal{O}^h_{X,x}$ is $n$.*

*Proof.* The equivalence of (1) and (2) follows from the fact that irreducible components of $U$ passing through $u$ are in 1-1 correspondence with minimal primes of the local ring of $U$ at $u$. The ring $\mathcal{O}_{X,x}$ is the henselization of $\mathcal{O}_{U,u}$, see discussion following Definition 0BGW. In particular (3) and (4) are equivalent by More on Algebra, Lemma 0BQ0. The equivalence of (2) and (3) follows from More on Morphisms, Lemma 0CB4. □